

CG&VR

MODELING

Modeling of the virtual world

1. Mapping of IO devices with the simulation scene.

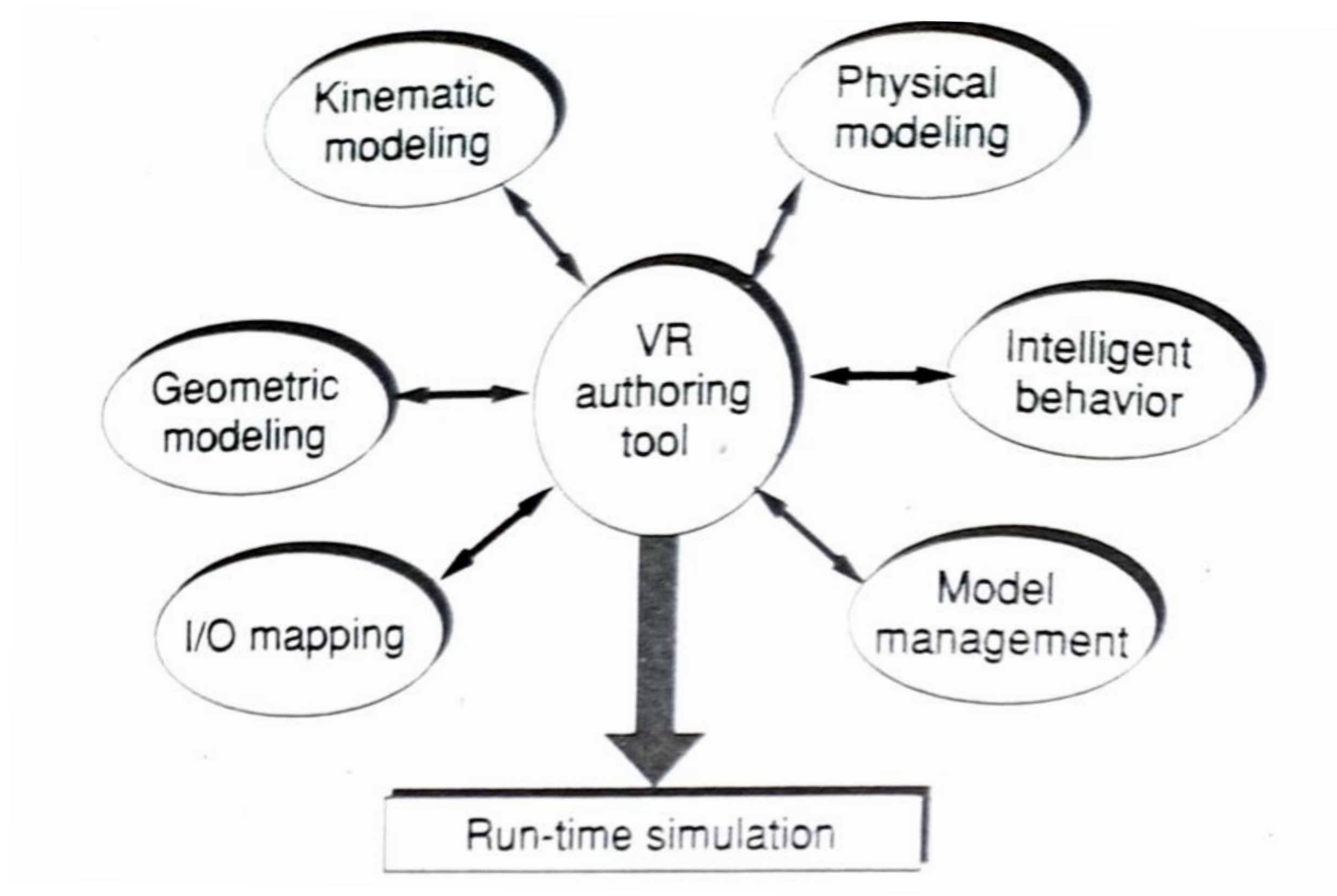
- For example, a sensing glove may control a virtual hand, or an HMD tracker may set the simulation view. VR interface mapping requires calibration to the user's characteristics as well as I/O communication coding.
- These tasks are usually solved by commercial toolkits.

2. The development of object databases in the virtual world.

- This means modeling object's shape, constraints, kinematic constraints, intelligent behaviour and physical characteristics (weight, Inertia, hardness etc)

3. Finally, in order to maintain a real time interaction with the simulation the model needs to be optimised during the model management step.

The VR modeling cycle



GEOMETRIC MODELING

- Geometric modeling describes the shape of virtual objects (polygons, triangles, vertices, splines) as well as their appearance (surface texture, surface illumination, and color)
- We must take care of 2 things
 - Virtual Object Shape
 - Object Visual Appearance

Virtual Object Shape

- The vast majority of virtual objects have their surface composed of **triangle meshes**.
- Triangle meshes are preferred because they **use shared vertices and are faster to render**. The vertices of shared edges are the same for adjacent triangles.
- Model using triangle meshes will need **less memory to be stored and will be loaded faster** by the application stage of the rendering pipeline
- The **more intricate the details** needed in the curved surface are, **the more triangles are needed**.
- An **alternative way** of shape representation is using **parametric surfaces**.

Methods by which object surfaces can be constructed:

1. Using a toolkit editor:

- Text based specifications of vertex coordinates and connectivity is necessary to create the shape and normals need to be specified if smooth shading iOS required.
- Amount of time required is inversely proportional to the programmer's skills.
- Very expensive if the virtual scene has extremely complex models.

2. Importing CAD files:

- Standard programs like AutoCAD or 3-D studio can be used.
- Pre existing models of mechanical assemblies and buildings can be used to populate the virtual world.
- Once the AutoCAD object files are saved they need to be converted into formats compatible with the particular VR toolkit used to run the real time simulation.

3. Creating surfaces with a 3D digitizer:

- Used when the virtual object required for the simulation is not a part of an existing database.
- Example- Immersion MicroScribe 3D mechanical arm.



Immersion MicroScribe 3D mechanical arm

The process of digitizing a clay model of the object starts with calibration of the MicroScribe.

This is done by selecting three points in space and mapping them with three points on the computer screen selected using a mouse.

This process establishes the distance scaling as well as the orientation of the model in relation to the virtual space.

The user needs to trace a mesh on the real object surface and select desired vertices by pressing the Micro Scribe pedal.

Symmetrical objects have the advantage that the manual digitization process needs to be done only for half of their surface. The other half is obtained by mirroring the data already input in the computer and stitching the two halves together.

4. 3D Scanners:

- Specialized software drives the laser probe above the surface of the object. The laser probe projects a line of laser light onto the surface while 2 sensor cameras continuously record the changing distance and shape of the laser line in three dimensions (XYZ) as it sweeps along the object.

5. Using online 3D object databases:

- This involves purchasing already created 3D models from various commercial databases.
- Such models usually have three levels of detail, low (L), medium (M), and high (H), according to the number of polygons in the model.
- Each model comes with two data files, one containing vertex coordinates and the other for vertex connectivity.
- Purchasing 3D models is fast, but can be expensive.
- The choice is limited to the database contents which are by necessity finite.

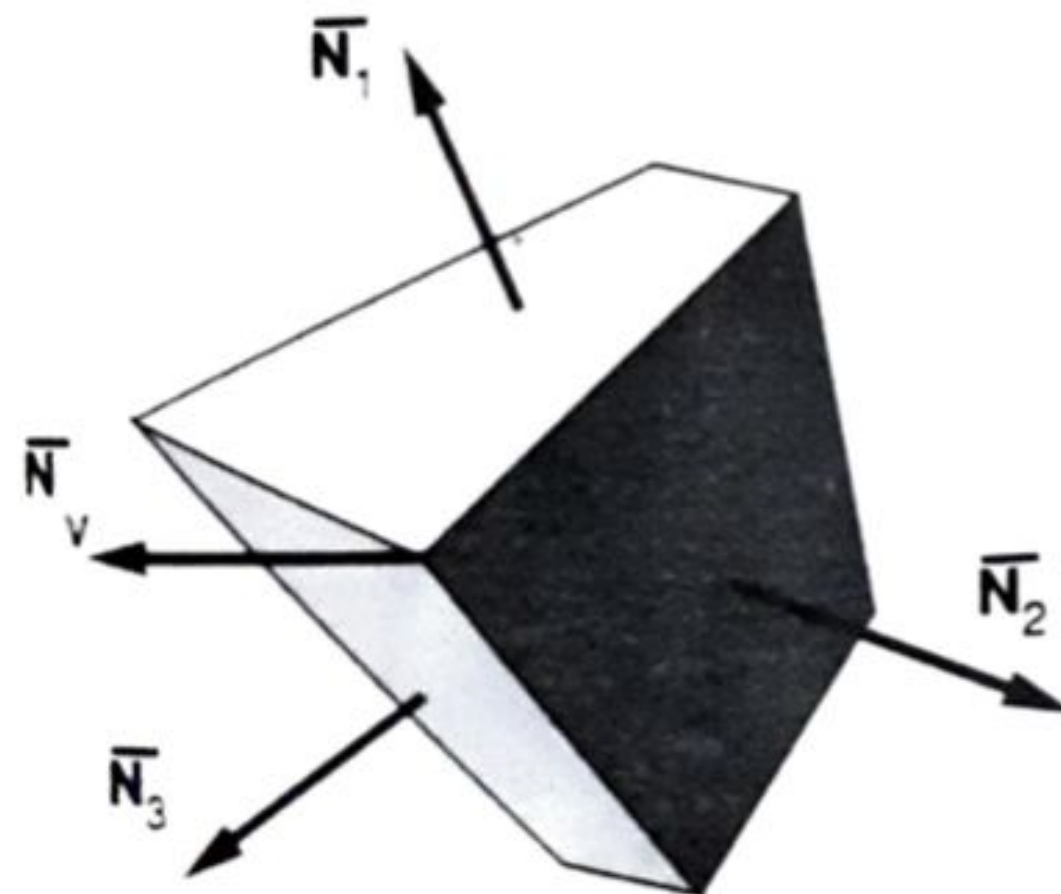
Object visual appearance

- The appearance of the object will strongly depend on the **type and placement of virtual light source** as well as on the **object's surface reflectivity coefficient**.
- Another factor determining surface complexity is the **surface texture**.

1. Scene Illumination:

- **Local scene illumination** treats the **interactions** between **objects** and **light sources** **in isolation neglecting the interdependences between objects.**
- **Global illumination** models the **inter reflections between objects and shadows**, resulting in a **more realistic-looking scene.**
- A **common local illumination method** which shades objects based on light intensity interpolation is **Gouraud shading.**
- This shading mode requires that the **normal of mesh** at each point on the surface of the **virtual object** **is a polygonal**
- The **normal**, N_v , is given by

$$\overline{N_v} = \frac{\sum_{i=1}^n \overline{N_i}}{|\sum_{i=1}^n \overline{N_i}|}$$



2. Texture mapping

- Texture mapping is a technique **performed in the rasterizing stage** of the graphics pipeline in order to **modify** the object model's surface properties such as **color, specular reflection or pixel normals**

THANK YOU